EAT40005 Engineering Technology Projects A

Individual Work Log

PROJECT NAME:	Whole-Body Model Predictive Control for Industrial Humanoids
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Cumulative Summary of week 10

Week `10	Major work done this week	Total hours spent this week	Cumulative Total
1	**Improved humanoid standing stability through iterative PID/PD balance tuning**	**4**	10
2	**Refined crouched humanoid posture and COM alignment for improved balance geometry**	5	11
3	**Implemented torso stabilization and pelvis alignment control systems**	3	7
4	**Improved actuator mapping, joint locking and humanoid control structure**	**3**	**9**
5	**Implemented startup stabilization and simulation reset handling system**	**2**	**9**
6	**Conducted humanoid balance testing, instability analysis and controller debugging**	**1.5**	**3**
7	**Researched NVIDIA Isaac Sim, ROS2 humanoid workflows and RL locomotion systems**	**3**	**5**
8	**Planned recovery stepping and future locomotion transition architecture**	**2**	**11**

Week 1

WEEK #	1		
Dates covered:	**02/03/2026 – 08/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Researching on potential projects	Completed	**10**	**Asking for personal project proposal**
Communicating with the convenor	completed	**4**	**Choosing a given project from the list**
TOTAL WEEKLY TIME SPENT		**14**	

Summary/weekly reflection for Week 1:

- Researched on available projects to find the suitable one
- Communicating with the convenor for support

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
More research into available projects	15/03/2026
osing a suitable project	15/03/2026

Week 2

WEEK #	2		
Dates covered:	**09/03/2026 – 15/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Researching on projects	Completed	**8**	**Chosing the humanoid robot control project**
Researching on the selected project	Completed	**12**	**Drafting a plan**
TOTAL WEEKLY TIME SPENT		**20**	

Summary/weekly reflection for Week 2:

- Researching on available projects
- Researching on the selected project

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Finding a Teammate	22/03/2026

Finishing up the project and team plan assignment	22/03/2026

Week 3

WEEK #	**3**		
Dates covered:	**16/-3/2026 – 22/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Project allocation and understanding project requirements	Completed	**2**	**Task devision among members**
Research on simulation tools (PyBullet / MuJoCo)	Completed	**5**	**Installed necessary libraries**
Finshing assignment 1	Completed	8	Submitted the assignment
TOTAL WEEKLY TIME SPENT		**15**	

Summary/weekly reflection for Week 3:

- Project was assigned and understood
- Team roles and responsibilities were defined
- Initial research on humanoid robotics and control systems conducted
- Completed major sections of the project proposal/report
- Set up tools for development (GitHub, communication platforms)

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	**EXPECTED COMPLETION**

Working on Assignment 2 (research report)	29/03/2026
Researching and start working on the project implementation	On going

Week 4

WEEK 4	**4**		
Dates covered:	**23/03/2026 – 29/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Completed research document	Completed	5	Research doc submission
Continued research on Model Predictive Control (MPC)	On going	**4**	drafts
Research on inverted pendulum model	On going	4	drafts
Planning system architecture	Completed	**3**	Finalised System Architecture
TOTAL WEEKLY TIME SPENT		**16**	

Summary/weekly reflection for Week #:

- Conducted in-depth research on MPC and control systems
- Identified inverted pendulum as the correct modelling approach
- Defined high-level system architecture
- Started research report writing
- Set up Jira sprint structure

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Begin implementation of inverted pendulum simulation	05/04/2026
Testing simulation Behavior	05/04/2026

Week 5

WEEK #	5		
Dates covered:	**30/03/2026 - 12/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Studied PID control fundamentals	Completed	3	Understood feedback control and tuning parameters
Started inverted pendulum simulation implementation	Completed	4	Implemented basic system model in Python
Tested simulation behaviour	Completed	2	Verified system response under simple conditions
Further research on MPC implementation methods	Completed	2	Focused on optimisation-based control
TOTAL WEEKLY TIME SPENT		11	

Summary/weekly reflection for Week #:

- *Studied PID control as baseline method*
- *Started implementation of inverted pendulum simulation*
- *Conducted initial testing of system behaviour*
- Difference between reactive (PID) and predictive (MPC) control
- Importance of feedback loops in maintaining system stability
- Simulation helps validate concepts before complex implementation

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Working on Sprint report 1	19/04/2026
Implementing and testing on PID and MPC	19/04/2026

Week 6

WEEK #	6		
Dates covered:	**12 /04/2026 – 19/04/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Implemented PID controller with full cart-pole dynamics	Completed	5	Integrated PID with realistic physics (cart + pole interaction)
Implemented MPC controller with prediction and cost function	Completed	4	Developed simulate_step() and cost-based optimisation
Fixed instability issues in simulation	Completed	2	Resolved issues like drifting, unrealistic motion, and imbalance
Implemented data logging and performance graphs	Completed	2	Generated theta, position, and force graphs
TOTAL WEEKLY TIME SPENT		11	

Summary/weekly reflection for Week #:

- Successfully implemented both PID and MPC controllers using realistic cart-pole dynamics
- Understood how system states (position, velocity, angle) are interconnected
- Learned that controller performance depends heavily on tuning (gains for PID, cost function for MPC)
- Observed differences between PID (reactive) and MPC (predictive) in handling disturbances
- Identified issues such as drift and continuous movement due to lack of position constraints

- Verified that the cart moves in response to pole angle, confirming correct physical coupling
- Introduced disturbance testing to simulate real-world conditions

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Plan and organise Sprint 2	26/04/2026
Implementing and testing MPC humanoid simulations	26/04/2026

Week 7

WEEK #	7		
Dates covered:	**19 /04/2026 – 26/05/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Implemented PID controller with full cart-pole dynamics	Completed	5	Developed base simulation using PyBullet
Implemented MPC controller with prediction and cost function	Completed	4	Resolved Not connected to physics server error
Fixed instability issues in simulation	Completed	2	Verified joint control and motion responses
Implemented data logging and performance graphs	Completed	2	Improved modular structure for scalability
TOTAL WEEKLY TIME SPENT		13	

Summary/weekly reflection for Week #:

- Focused on implementing the core simulation environment
- Successfully resolved critical PyBullet connection issues
- Tested initial humanoid robot movement and control
- Improved overall code structure and modularity
- Gained better understanding of physics-based simulation challenges

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Integrating AI control logic (basic decision-making)	03/05/2026
Enhancing robot movement with smoother transitions	03/05/2026
Begin implementing environment interaction (obstacles/tasks)	03/05/2026

Week 8

WEEK #	7		
Dates covered:	**26 /04/2026 – 03/05/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Migrated project focus from PyBullet humanoid to MuJoCo simulation environment	Completed	5	Reviewed updated client requirements and selected MuJoCo for locomotion/object interaction tasks
Researched MuJoCo architecture, XML models, actuators, sensors and control loop structure	Completed	4	Investigated MuJoCo workflow for humanoid robotics and simplified model control
Designed simplified humanoid robot structure with reduced degrees of freedom	Completed	2	Planned leg joints, torso, arm joints and gripper for efficient implementation
Planned locomotion system for robot to move toward target object	Completed	2	Developed walking logic using scripted gait / control state transitions
Planned object interaction sequence (reach, grasp, lift)	Completed	3	Defined task stages for navigation and manipulation behaviour
Updated modular project architecture for MuJoCo implementation	Completed	2	Refined folders for models, controllers, tasks and utilities
TOTAL WEEKLY TIME SPENT		13	

Summary/weekly reflection for Week #:

- Project direction transitioned from balancing-only control to full humanoid task execution in MuJoCo.
- Gained strong understanding of MuJoCo simulation workflow, XML robot modelling and actuator control systems.
- Planned a simplified humanoid robot to reduce complexity while still demonstrating locomotion and manipulation.
- Designed behaviour flow for robot to walk to a target, stop accurately, reach an object and lift it.
- Improved understanding of how previous PID/MPC knowledge transfers into movement and interaction tasks.

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Integrating AI control logic (basic decision-making)	03/05/2026
Enhancing robot movement with smoother transitions	03/05/2026
Begin implementing environment interaction (obstacles/tasks)	03/05/2026

Week 9

WEEK #	9		
Dates covered:	**10 /04/2026 – 17/05/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Migrated humanoid robotics project from PyBullet to MuJoCo simulation environment	Completed	5	Evaluated limitations of PyBullet for humanoid balancing and locomotion tasks. Migrated project to MuJoCo due to improved articulated-body physics, actuator handling and humanoid simulation accuracy. Configured MuJoCo Python environment and verified robot model loading through MJCF scene files.
Researched MuJoCo humanoid control workflow, XML structures and actuator systems	Completed	4	Investigated MuJoCo architecture including XML robot definitions, body hierarchies, joints, actuators,

			sensors and simulation stepping. Studied how joint indexing, qpos/qvel states and actuator controls interact within humanoid systems.
Integrated Unitree H1 humanoid model into MuJoCo environment	Completed	2	Loaded and configured Unitree H1 humanoid model into MuJoCo simulation. Verified articulated joints, actuator mappings and simulation rendering. Debugged XML include issues and validated humanoid movement within the environment.
Implemented custom PID-based humanoid balancing controller	Completed	2	Developed joint-level PID/PD control system for hip, knee and ankle stabilization. Implemented posture tracking and balance correction logic to maintain humanoid crouched standing pose. Tuned stiffness and damping values iteratively through simulation testing.
Developed simplified MPC-inspired predictive balancing module	Completed	3	Implemented lightweight MPC-style balance prediction system to estimate future pitch drift and generate corrective stabilization bias. Combined MPC outputs with lower-level PD balancing system.
Implemented center-of-mass (COM) estimation and stabilization logic	Completed	2	Created simplified COM approximation module using robot base position and velocity data. Used COM feedback to improve humanoid balance and reduce forward/backward drift during standing.
TOTAL WEEKLY TIME SPENT		13	

Summary/weekly reflection for Week #:

- The project successfully transitioned from PyBullet to MuJoCo to improve humanoid simulation realism, articulated-body dynamics and balancing behaviour.
- A custom humanoid balancing framework was developed for the Unitree H1 robot using MuJoCo physics simulation and Python-based control systems.
- Significant understanding was gained regarding humanoid kinematics, actuator behaviour, balance dynamics and center-of-mass stabilization.
- Implemented modular balancing architecture including:
    - PID/PD joint stabilization

    - MPC-inspired predictive balancing
    - COM estimation
    - posture stabilization
    - torso alignment control
- Through iterative tuning and posture optimization, the humanoid robot progressed from immediate instability to maintaining a semi-stable crouched standing posture for approximately 4–5 seconds before balance drift caused collapse.
    - Multiple important humanoid robotics challenges were identified during testing, including:
        - pelvis yaw misalignment
        - torso drift
        - knee extension instability
        - ankle overcompensation
        - support polygon limitations
    - Research into NVIDIA Isaac Sim, ROS2 humanoid workflows and reinforcement learning locomotion systems improved understanding of modern humanoid robotics pipelines and future project directions.

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Improve humanoid standing stability through advanced balance tuning	24/05/2026
Begin implementing recovery stepping behaviour for fall prevention	24/05/2026
Improve COM estimation and torso stabilization systemimplementing environment interaction (obstacles/tasks)	24/05/2026
Research reinforcement learning approaches for humanoid locomotion	24/05/2026
Begin environment interaction and target-navigation behaviour	24/05/2026

Week 10

WEEK #	10		
Dates covered:	17/05/2026 – 24/05/2026		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Improved humanoid standing stability through iterative balance tuning	Completed	5	Tuned proportional and derivative gains for hip, knee and ankle joints to improve crouched standing stability. Reduced oscillations and improved short-duration balance behaviour.
Refined crouched humanoid posture and COM alignment	Completed	4	Experimented with different hip, knee and ankle angles to align the robot's center of mass above the support region of the feet. Identified crouched stance as significantly more stable than upright posture.
Implemented torso stabilization and pelvis alignment control	Completed	3	Investigated torso and pelvis rotational instability. Added torso stabilization logic and yaw alignment control to reduce body twisting and improve structural posture consistency.
Improved actuator and joint control structure	Completed	3	Refined actuator indexing and control application for humanoid joints. Locked unused degrees of freedom including arm joints and lateral joints to isolate sagittal balancing behaviour.
Implemented startup stabilization and reset handling system	Completed	2	Added initial pose injection and startup stabilization phase to allow robot to settle into nominal posture before active balancing began. Implemented automatic pose reset after simulation restart.
Conducted humanoid balance testing and debugging sessions	Completed	4	Tested multiple controller configurations and analyzed instability behaviours including torso drift, knee extension, ankle collapse and COM overshoot. Iteratively refined control response

			based on observed simulation behaviour.
Researched humanoid locomotion pipelines using NVIDIA Isaac Sim and ROS2 workflows	Completed	3	Investigated modern humanoid robotics pipelines including Isaac Sim, ROS2 integration, IMU systems, reinforcement learning locomotion and humanoid state estimation workflows.
Planned recovery stepping and locomotion transition architecture	Completed	2	Designed next-stage architecture for recovery stepping and future locomotion control to prevent accumulated balance drift from causing collapse.
TOTAL WEEKLY TIME SPENT		26	

Summary/weekly reflection for Week #:

- Significant progress was made in improving humanoid balance stability within the MuJoCo simulation environment.
- The humanoid robot successfully transitioned from immediate uncontrolled collapse to maintaining a semi-stable crouched standing posture for approximately 4–5 seconds using custom balancing logic.
- One of the major findings during this sprint was that humanoid stability depended heavily on posture geometry and center-of-mass alignment rather than only controller strength.
- The project identified several key humanoid balancing challenges including:
    - pelvis yaw misalignment
    - torso rotational drift
    - knee extension instability
    - ankle overcompensation
    - accumulated COM drift
- Improved understanding was gained regarding:
    - humanoid kinematics
    - articulated-body dynamics
    - support polygon limitations
    - predictive balancing
    - multi-joint stabilization systems
- Additional research into NVIDIA Isaac Sim, ROS2 and reinforcement learning humanoid control pipelines helped establish a clearer long-term direction for future humanoid locomotion development.
- The balancing framework now behaves as an active stabilization system rather than an uncontrolled physics simulation, representing a major improvement over earlier iterations.

Plan for the nxt week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Implement recovery stepping behaviour for fall prevention	31/05/2026
Improve predictive balancing using enhanced MPC logic	31/05/2026
Begin locomotion and directional walking experiments	31/05/2026
Improve COM estimation using additional body-state information	31/05/2026
Investigate reinforcement learning integration for humanoid control	31/05/2026